

1 Basic Knowledges

Completeness Axiom: If $\exists \text{ UP} \Rightarrow \text{LUB} (\sup)$; If $\exists \text{ LB} \Rightarrow \text{GLB} (\inf)$.

Archimedean Property: $\forall x \in \mathbb{R}^+ \Rightarrow \exists n \in \mathbb{N} \text{ s.t. } n > x$.

Eventually True: P_n is eventually true iff $\exists$ an index N s.t. P_n true for $n > N$.

Important Inequation: $|x + y| \leq |x| + |y|$; $|x - y| \geq |x| - |y|$; $||x| - |y|| \leq |x \pm y|$.

Bernoulli's Inequality: If $\alpha > 0, \delta \geq -1$. Then (1) $(1 + \delta)^\alpha \leq 1 + \delta\alpha, \alpha \in (0, 1]$; (2) $(1 + \delta)^\alpha \geq 1 + \delta\alpha, \alpha \in [1, +\infty)$

Injective Surjective: Let $f : S \rightarrow T$: If **1-1** : $|S| \leq |T|$; If **onto** : $|S| \geq |T|$; $|S|^{|T|}$ maps for $S \rightarrow T$.

· If S and T be two finite sets. Then $|S \times T| = |S||T|$ · f has inverse iff it is bijection.

Countable: S is countable if $\exists$ bijective $f : \mathbb{N} \rightarrow S$; If S is finite $\Rightarrow$ at most countable.

· If S_1, S_2 is countable $\Rightarrow S_1 \cup S_2$ is countable and $S_1 \times S_2$ is countable. ps: $\mathbb{R}$ is uncountable.

Converge Sequence: If sequence is convergent $\Rightarrow$ it is bounded.

Permutation: $f = (a_1 a_2 a_3 \cdots a_k) = (a_1 a_2)(a_2 a_3) \cdots (a_{k-1} a_k) = (a_1 a_k)(a_1 a_{k-1}) \cdots (a_1 a_2)$; $f^{-1} = (a_k a_{k-1} \cdots a_1)$.

· If $f = gh$ then $f^{-1} = h^{-1}g^{-1}$

Monotone Convergent Theorem: 单调有界必收敛。

2 Analysis

2.1 Sequence Limit

Def of Seq Limit: $(x_n)_{n \in \mathbb{N}}$ is convergent to a iff $\forall \varepsilon > 0, \exists N > 0$ s.t. $|x_n - a| < \varepsilon$ for all $n > N$

· If $\lim x_n$ exist then so does $\lim |x_n| = |\lim x_n|$ ps: $\lim x_n = 0 \Leftrightarrow \lim |x_n| = 0$

Operation in Seq Limit: Squeeze Theorem ; Monotone Convergence Theorem ; $\lim \alpha x_n = \alpha \lim x_n$

· $\lim(x_n + y_n) = \lim x_n + \lim y_n$; $\lim x_n \cdot y_n = \lim x_n \cdot \lim y_n$; $\lim \frac{x_n}{y_n} = \frac{\lim x_n}{\lim y_n}, y_n \neq 0$

· **Comparison Theorem (For Seq):** If $\lim x_n, y_n$ exist, and $x_n \leq y_n$ for all $n \geq N$ then $\lim x_n \leq \lim y_n$.

Useful Seq Limit: $\lim \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Theorem: 有限上/下界的区间内, 一定有一收敛于上/下界的数列。

Theorem: For $(x_n)_{n \in \mathbb{N}}$, if it's unbounded increasing $\Rightarrow \lim x_n = +\infty$ 同理 unbounded decreasing.

Def of Seq Divergence: $(x_n)_{n \in \mathbb{N}}$ is divergent to $+\infty$ iff $\forall M > 0, \exists N$ s.t. $x_n > M$ for all $n > N$ ps: 同理 $-\infty$.

2.2 Subsequences

Def of Subsequences: (x_{n_k}) where $n_1 < n_2 < n_3 < \cdots$ is an increasing sequence of natural numbers.

Subsequence Theorem: In each case, the subsequence can be taken to be monotonic.

1. $\exists a \in \mathbb{R} \text{ s.t. } \forall \varepsilon > 0$, there are infinitely many $n \in \mathbb{N} \text{ s.t. } |x_n - a| < \varepsilon \Leftrightarrow \exists$ subsequence of $(x_n) \text{ s.t. } \lim x_{n_k} = a$

2. $(x_n)_{n \in \mathbb{N}}$ is not bounded above (below) $\Leftrightarrow \exists$ subsequence of $x_n \text{ s.t. } \lim x_{n_k} = \infty (-\infty)$

3. If $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

Bolzano-Weierstrass: $\forall$ bounded sequence (real number) has a convergent subsequence.

2.3 Infinite Series

Def of Infinite Series: Let $s_n = \sum_{k=1}^n a_k$, then Infinite series $\sum_{k=1}^{\infty} a_k = \lim s_n$

Operation in Infinite Series: $\sum(a_n + b_n) = \sum a_n + \sum b_n$ and $\sum(\alpha a_n) = \alpha \sum a_n$

· **p-Test:** $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test** $|a_n|^{1/n}, |a_{n+1}/a_n|$ **Geometric Series:** $\sum p^n$ converges iff $|p| < 1$

· If $a_n \geq 0$, $\sum a_n$ converges iff its partial sum is bounded. ($|\sum_{k=1}^n a_k| < M$) ps: 就是用了 MCT

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n$, $\sum b_n$ convergent $\Rightarrow \sum a_n$ also ; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test:** If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0 \Rightarrow$ If $L \in (0, \infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.

· If $L = 0$: $\sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty$: $\sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f : [1, \infty) \rightarrow \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^{\infty} f(x)dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum (-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow [\sum a_n \text{ convergent} \Leftrightarrow \sum 2^n a_{2^n} \text{ also}]$

Cauchy Product: If $\sum a_n$ convergent abs, $\sum b_n$ convergent $\Rightarrow (\sum a_n)(\sum b_n) = \sum c_n, c_n = \sum_{i=1}^n a_i \times \sum_{i=1}^n b_i = \sum_{i=1}^n a_i b_{n-i}$

· By Cauchy product $e^{x+y} = e^x e^y$ and $\log(xy) = \log(x) + \log(y)$ $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ $\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}} = e$ q

Cauchy's Criterion (Sequence): $\sum a_n$ convergent iff $\forall \varepsilon > 0, \exists N$ s.t. $m \geq n \geq N, |a_n + \dots + a_m| < \varepsilon$

2.4 Continuity

Def of Continuity: Let $f : A \rightarrow B, f$ continuous at x_0 iff:

1. **1st Def:** $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x \in A, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. ps: x 可以等于 x_0

2. **2nd Def:** $\forall (x_n)_{n \in \mathbb{N}} \in A, \text{ if } \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$ ps: x 可以等于 x_0

One-side Continuity: Let $f : A \rightarrow B$ continuous from left (or right) at x_0 iff: (也可以用 sequence 来定义)

$\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x \in A, |x - x_0| < \delta, \text{ and } x \leq x_0 \text{ (or } x \geq x_0) \Rightarrow |f(x) - f(x_0)| < \varepsilon$

Operation in Continuity: If f is continuous at x_0 : so does $f + g, \frac{f}{g}, fg$ (就单纯的乘), $f \circ g$.

· All polynomial and rational functions (coefficient $\in \mathbb{R}$) are continuous functions.

Sign Preserving Property: If g continuous at $x_0, g(x_0) \neq 0$

$\Rightarrow \exists \delta > 0$ s.t. $\forall x, |x - x_0| < \delta, \text{ sign}(g(x)) = \text{sign}(g(x_0))$ Moreover, $|g(x)| > \frac{1}{2}|g(x_0)|$

Theorem: If $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function, then f is closed and bounded.

The Extreme Value Theorem: If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then f has a max and min value.

Bolzano Theorem: If $f : [a, b] \rightarrow \mathbb{R}$ is continuous with $f(a)f(b) < 0 \Rightarrow \exists c \in (a, b)$ s.t. $f(c) = 0$

Intermediate Value Theorem: $f : [a, b] \rightarrow \mathbb{R}$ is continuous, $f(a) \neq f(b), y \in [f(a), f(b)] \Rightarrow \exists c \in (a, b)$ s.t. $f(c) = y$

· If $f : [a, b] \rightarrow [a, b]$ is continuous. $\Rightarrow \exists c \in (a, b)$ s.t. $f(c) = c$

Theorem If $f : \mathbb{I} \rightarrow \mathbb{R}$ is continuous and injective. Then f is strictly monotone and its inverse also.

Theorem If $f : \text{Interval} \rightarrow \text{Interval}$. is strictly increasing, then f is continuous on it.

2.5 Limits of Functions

Def of Limit Point: x_0 is a limit point of A iff:

1. **1st Def:** $\forall \delta > 0, \exists x, x \neq x_0$ s.t. $|x - x_0| < \delta$

2. **2nd Def:** $\exists (x_n)_{n \in \mathbb{N}}, x_n \neq x_0$ s.t. $\lim x_n = x_0$

Def of Function Limit: $\lim_{x \rightarrow x_0} f(x) = L$ iff

1. **1st Def:** $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x \in A, |x - x_0| < \delta, x \neq x_0 \Rightarrow |f(x) - L| < \varepsilon$.

2. **2nd Def:** $\forall x_n, x_n \neq x_0$ s.t. $\lim_{n \rightarrow \infty} (x_n) = x_0$, then $\lim_{n \rightarrow \infty} f(x_n) = L$.

Operation in function limit: 和 continuity 一样 + Squeeze theorem ps: 同理也有 Sign Preserving Property.

· $(+\infty) + (+\infty) = +\infty$ $x \pm (\infty) = \pm \infty$ $x \cdot (\pm \infty) = \pm \infty$ $(+\infty)(-\infty) = -\infty$

· If $\lim f(x) = L, g$ is continuous at L , then $\lim g(f(x)) = g(L)$

Limit as $x \rightarrow +\infty$: $\lim_{x \rightarrow +\infty} f(x) = L$ iff $\forall \varepsilon > 0, \exists M$ s.t. $x > M$ we have $|f(x) - L| < \varepsilon$

Limit as $f(x) \rightarrow +\infty$: $\lim_{x \rightarrow x_0} f(x) = \infty$ iff $\forall M > 0, \exists \delta$ s.t. $\forall 0 < |x - x_0| < \delta$, we have $f(x) > M$

One / Two-sided Limit: **Two:** $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a, x \in I \setminus \{a\}} f(x)$ **One(Left):** $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a, x \in I \setminus (a, +\infty)} f(x)$

· $\lim_{x \rightarrow a^+} f(x) = L$ iff $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall a < x < a + \delta, |f(x) - L| < \varepsilon$

· $\lim_{x \rightarrow a} f(x)$ exists iff $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$

Theorem: f is continuous at a iff $\lim_{x \rightarrow a} f(x) = f(a)$

Comparison Theorem (for func): If $f(x) \leq g(x)$ (or $f(x) < g(x)$), for all $x \in I \setminus \{a\}$, then $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$

Isolated Point: x_0 is an isolated point of A iff $(x_0 - \delta, x_0 + \delta) \cap A = \{x_0\}$ ps: x_0 点独立

· If x_0 is an isolated point of A , then f is continuous at x_0 . ps: But $\lim_{x \rightarrow x_0} f(x)$ doesn't make sense.

Def of Removable and non-Removable Discontinuities:

1. **Continuous:** $\lim_{x \rightarrow x_0} f(x) = f(x_0)$

2. **Removable:** $\lim_{x \rightarrow x_0} f(x) = L \neq f(x_0)$

3. **non-Removable:** $\lim_{x \rightarrow x_0} f(x)$ not exist.

· ps: $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$

2.6 Differentiability on $\mathbb{R}$

Def of Derivative: A real function f is differentiable at point $a \in \mathbb{R}$ if:

1. **1st Def:** $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exist. ps: Difference quotient at x_0 : $Q(x) = \frac{f(x) - f(x_0)}{x - x_0}$
2. **2nd Def:** $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exist.
3. **Higher derivative Def:** $f^{(n+1)}(a) = (f^{(n)}(a))'$

Linear Approximation: Linear approximation of f near x_0 is: $f(x_0) + f'(x_0)(x - x_0)$

Theorem: If f is differentiable at a , then f is continuous at a .

Def fo Differentiable (On I): Let I be a non-degenerate interval. (即不是一个点)

1. $f : I \rightarrow \mathbb{R}$ is said to be differentiable on I if $f'_I(a) := \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exist for $\forall a \in I$
ps: 如果不在 endpoint $f'_I = f'$, 在端点定义: $\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$ and $\lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$

2. f is continuously differentiable := f differentiable + f' continuous on I .

Def of C^n : $C^n(I)$, $n \in \mathbb{N} \cup \{0\}$:= functions which (1) n -th derivative exists (2) derivative continuous on I .

$\cdot C^\infty(I) = \bigcap_{n \in \mathbb{N} \cup \{0\}} C^n$ and $C^m \subset C^n$ if $m > n$

Fermat Theorem: f differentiable at open interval, f has a max (min) point at $c \Rightarrow f'(c) = 0$ ps: Critical point: $f'(x) = 0$

Rolle's Theorem: If f continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$ Then: $\exists c \in (a, b)$ s.t. $f'(c) = 0$

Mean Value Theorem (MVT): Let f, g continuous on $[a, b]$ and differentiable on (a, b) : ps: $a=b$ 时依然成立

1. $\exists c \in (a, b)$ s.t. $\frac{f(b) - f(a)}{b - a} = f'(c)$ ps: Lagrange's Mean Value Theorem
2. $\exists c \in (a, b)$ s.t. $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$ ps: Cauchy's Mean Value Theorem (参数方程的 Mean value Theorem)

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c$ s.t. $f'(c) = m$

Corollary: If f differentiable in I and $f'(x) \neq 0$, then f' has constant sign.

Monotone Theorem: Let f continuous on $[a, b]$ and differentiable on (a, b) :

1. If $\forall x \in (a, b), f'(x) > 0 \Rightarrow f$ is strictly increasing on $[a, b]$
2. If $\forall x \in (a, b), f'(x) < 0 \Rightarrow f$ is strictly decreasing on $[a, b]$
3. If $\forall x \in (a, b), f'(x) = 0 \Leftrightarrow f$ is constant on $[a, b]$

Theorem: If f is 1-1 and continuous $\Rightarrow f$ is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

- 1. $f(I)$ is open interval.
- 2. f is invertible.
- 3. f^{-1} is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in $I, f'(x_0) = 0$: 1. $f''(x_0) > 0 (< 0)$ local min (max) point.

2.7 L'Hospital's Rule and Taylor's Theorem

L'Hospital's Rule: If f, g differentiable on $I \setminus \{a\}$ and $g(x), g'(x) \neq 0$ for $\forall x \in I \setminus \{a\}$:

If $\lim_{x \rightarrow a} \frac{\infty}{\infty}$ or $\lim_{x \rightarrow a} \frac{0}{0}$. Then: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$

Taylor's Theorem: If $f : (a, b) \rightarrow \mathbb{R}$ is differentiable function at $x_0 \in (a, b)$:

1. $P_n^{f, x_0}(x) = f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$ (Taylor's polynomial of degree n at x_0)
2. If $f^{(n+1)}$ exists on (a, b) , then: $\exists c \in (x, x_0)$ s.t. $f(x) = P_n^{f, x_0}(x) + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$

Special Example: $f(x) = e^{-\frac{1}{x}}, x > 0; 0, x \leq 0$ ($f^{(n)} = 0$ for all n , and $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = 0 \neq f(x)$)

Special Example: $f(x) = \sum_{n=0}^{\infty} e^{-n} \cos(n^2 x)$ (A Taylor Series that converges only at the center $x_0 = 0$)

Trigonometric function: we define function function sin, cos by:

1. $\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \dots$
2. $\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \dots$

The Taylor Series of f does not always converge to f . e.g. $f(x) = e^{-1/x}, x > 0; 0, x \leq 0$

3 Group

3.1 Graphs

Def of Graphs Γ : A finite set of vertices joined by edges. **Valency** is number of edges emerging from a vertex.

Isomorphism (for Graphs): $f : \Gamma_1 \rightarrow \Gamma_2$ is isomorphism iff it is a bijection. ps:vertex 对应 edge 数一样; 不唯一.

Symmetry of Graph: It's a isomorphism from the graph to itself. (Always Faithful)

· If $\Gamma_1 \cong \Gamma_2$ then $Sym(\Gamma_1 \cup \Gamma_2) = Sym(\Gamma_1)Sym(\Gamma_2)$

3.2 Group and Basic Groups

Binary Operation ($\odot$): A binary operation $\odot$ on set A is: $\forall a, b \in A$ then $a \odot b \in A$ and unique.

Def of Group ($G, \odot$): **G1** Closure: $\forall a, b \in G$, then $a \odot b \in G$ **G2** Associativity: $(a \odot b) \odot c = a \odot (b \odot c)$

G3 Identity: $\exists e \in G$ s.t. $e \odot g = g \odot e = g$ **G4** inverse: $\forall g \in G$ there $\exists g^{-1} \in G$

· **Order of G** $|G|$: The # of elements.

ps: $|G \times H| = |G||H|$

· **Order of elements in G** $o(g)$: min # n s.t. $g^n = e$. ps: $o(g, h) = lcm(o(g), o(h))$

· **Abelian Group**: If G is a group and $\forall g, h \in G$, we have $g \odot h = h \odot g$.

· The set of symmetries of a graph forms a group. If G, H are groups, then $G \times H$ also.

Cyclic Group $\langle g \rangle$: If $G = \{..., g^{-2}, g^{-1}, e, g^1, g^2, ...\}$

Dihedral Group (D_n): The group of symmetries of a regular n -sided polygon. $|D_n| = 2n$ we have: $hg^k = g^{n-k}h$

Symmetric Group (S_n, S_X): All bijection from $\{1, 2, ..., n\}$ or set A to itself. $|S_n| = n!$

General Linear Group (GL_n): The $n \times n$ real matrix A , $det(A) \neq 0$.

· $SL_n: \{A \in GL_n \mid det(A) = \pm 1\}$ $SO_n: \{A \in SL_n \mid A^T = A^{-1}\}$ ps: SO_2 : 圆的 rotation 对称.

· **Orthogonal Group** (O_n): The $n \times n$ real matrix A , $A^{-1} = A^T$, $det(A) = \pm 1$ ps: O_2 : 圆的所有对称

3.3 More on Cyclic Group, Subgroups, Cosets, Lagrange's Theorem

Def of Subgroup: If $H \subseteq G$, H, G are group, then H is a subgroup of G . ps: Trivial subgroup: $\{e\}$

Def of Nontrivial / proper subgroup: A subgroup which is not Identity or G . / is not G

Subgroup Criterion: If $H \subseteq G$, G is group. **SG1** $H \neq \emptyset$ **SG2** $\forall g, h \in H$, $gh \in H$ **SG3** $\forall h \in H$, $h^{-1} \in H \Rightarrow H$ is subgroup.

· If H, K are two subgroups of G , then $H \cap K$ dose also.

Proposition of Cyclic Group: Let $G = \langle g \rangle$

1. Every cyclic group G is abelian. Every subgroup of G is cyclic.

2. G is a cyclic group of order n iff G has an element of order n .

3. If G and H are cyclic group, then $G \times H$ is cyclic iff $\gcd(|G|, |H|) = 1$

Def of Cosets: Let H be a subgroup of G , G is a group, $g \in G$, **Left Coset**: $gH = \{gh \mid h \in H\}$. **Right Coset**: Hg .

Proposition of Coset: Let H, K be subgroup of group G . **Index of H** $= |G/H| = \frac{|G|}{|H|}$

1. $\forall h \in H, hH = H$ $\forall g, h \in G$, If $h \in gH$, then $hH = gH$

2. There is a bijective function $\theta : H \rightarrow gH$ Thus, $|gH| = |H|$.

3. The set of Left Cosets of H in G called G/H e.g. $G = \mathbb{Z}_4, H = \{0, 2\}$ then $G/H = \{\{0, 2\}, \{1, 3\}\}$

If $\sim$ is an Equivalence Relation on a Set X , then X is partitioned into pieces (which is the Equivalence Classes)

Lagrange's Theorem: If G is a group, H is a subgroup of G . Then we have $|G| = m|H|$, $m \in \mathbb{Z}$.

· **Corollary**: Let $g \in G$, G is a group.

Fermat's Little Theorem: If p is prime, $p \nmid a$, then $a^{p-1} \equiv 1 \pmod p$

1. $o(g) \mid |G|$ $g^{|G|} = e$.

2. If $|G|$ is prime. Then G is cyclic group. and the subgroup of G is only $\{e\}$ and G

3. If $|G| < 6$, then G is abelian. ps: D_3 is the smallest non-abelian group.

3.4 Homomorphism and Isomorphism, Kernel, Image, Normal

Def of Homomorphism: A map $\phi : G \rightarrow H$ is homomorphism iff $\forall x, y \in G$ $\phi(xy) = \phi(x)\phi(y)$

· ps: Nonconstant homomorphism: $\phi(x) = c$ for all x .

Def of Isomorphism ($G \cong H$): If map $\phi : G \rightarrow H$ is homomorphism and bijective, then it's Isomorphism. ϕ^{-1} also.

Proposition of Isomorphism: Automorphism: $\phi : G \rightarrow G, G \cong G$

1. Every Cyclic group of order n (C_n) is isomorphism. $C_n \cong (\mathbb{Z}_n, +)$ $(\mathbb{R}, +) \cong (\mathbb{R}^*, \cdot)$

2. If $G \cong H$, G is abelian then H is abelian.

3. If $\phi : G \rightarrow H$, ϕ Homomorphism, then: $\phi(e_G) = e_H$ $\phi(g^{-1}) = (\phi(g))^{-1}$ If $\phi : 1-1$, then: $o(g) = o(\phi(g))$

Def of Ker/Im: Let $\phi : G \rightarrow H$ be homomorphism. $Im\phi := \{h \in H | h = \phi(g), g \in G\}$ $Ker\phi := \{g \in G, \phi(g) = e_H\}$
 $\cdot Ker\phi, Im\phi$ are both subgroup of G and H

Def of Normal ($H \triangleleft G$): A subgroup H of G is normal iff $\forall g \in G, gH = Hg$.

Proposition of Ker/Im/Normal: Let $\phi : G \rightarrow H$ be a group homomorphism.

Transitive	Faithful	Example
T	T	S_2 act on $\{1, 2\}$
T	F	S_2 act on $\{e\}$ by $g \cdot x = e, \forall g$
F	T	$G = Sym(\cdot - \cdot - \cdot)$ $X = \{1, 2, 3\}$
F	F	S_2 act on $\{1, 2\}$ by Trivial Action

1. $Ker\phi \triangleleft G$

2. ϕ is 1-1 iff $Ker\phi = \{e_G\}$ If ϕ is 1-1, then $G \cong Im\phi$

Theorem: Let H, K be subgroups of G and $H \cap K = \{e\}$ Then $|HK| = |H||K|$

1. Map $\phi : H \times K \rightarrow HK$ given by $\phi : (h, k) \rightarrow hk$ is bijective.

2. If $hk = kh$, then $HK \cong H \times K$, and HK is subgroup of G .

3.5 Group Action, Orbits, Stabilizers and its Theorem

Def of Action: If G is group, X is nonempty set. A (left) action of G on X is:

$\cdot$ A map: $G \times X \rightarrow X$, written $(g, x) \rightarrow g \cdot x$ s.t. $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x$ and $e \cdot x = x$

$\cdot$ ps: Trivial action: (action map:) $g \cdot x = x$ for $\forall g \in G$

Action of Group itself: 1. Left Action: $= g \cdot h = gh$ 2. Right Action: $= g \cdot h = hg^{-1}$ 3. Conjugate Action: $= g \cdot h = ghg^{-1}$

Stabilizer: Let G act on X , then $Stab_G(x) := \{g \in G | g \cdot x = x\}$ ps: It's always a subgroup of G .

Fix: Let G act on X , then $Fix(g) := \{x \in X, g \cdot x = x\}$

Kernel of Action ($N \triangleleft G$): $N := \{g \in G | \forall x, g \cdot x = x\} = \cap_{x \in X} Stab_G(x)$

Def of Faithful and Trivial: If Kernel = $\{e\}$ the action is faithful. If $f(x) = e$ for all $x \in X$, then it is trivial. $\downarrow$

Orbit of Action: Let G act on X , then $Orb_G(x) := \{g \cdot x | g \in G\}$ $\parallel$ The only member of G that fixes everything in X is identity.

Def of Transitive: G act on X is Transitive if $\forall x, y \in X, \exists g \in G$ s.t. $y = g \cdot x$. (X is a single orbit under G)

Theorem: Let G act on X , then $x \sim y \Leftrightarrow y = g \cdot x$ for some $g \in G$ def a equivalence relation on X .

$\cdot$ The equivalence classes are the orbits of G . (即可以得到 orbits 可以将 X partition)

Send Notation: Let G act on $X, x, y \in X$, then $Send_x(y) = \{g \in G | g \cdot x = y\} \subseteq G$

$\cdot$ Let G act on $X, x, y \in X$. If $\exists g \in G$ s.t. $y = g \cdot x$ (e.i. x, y are in the same orbit) then $send_x(y) = gStab_G(x)$

$\cdot$ 即: 所有让 x 回到 x 的 action $\in G$ 左乘一个让 x 到 y 的 action $\in G$ 可以使全部 g -action 都为 x 到 y 的 action

Theorem: $\phi : Orb_G(x) \rightarrow G/H, H = Stab_G(x)$ is bijection.

Orbit-Stabilizer Theorem: Let G act on X , then $|Orb_G(x)| \times |Stab_G(x)| = |G|$

Cauchy's Theorem: Let G be a group, p prime. If $p \mid |G|$ then G contains an element of order p .

3.6 Counting, Conjugacy and Class Equation

Polya Counting: # orbits of G in $X = \frac{1}{|G|} \sum_{g \in G} |Fix(g)|$

$\cdot$ **Coloring Problem:** Let G act on $X, X = \{\text{All coloring vertice}\}$. Then we have: # coloring = # orbits of G act on X .

Conjugation Action: Let G act on itself Then $= g \cdot h = ghg^{-1}$ The **orbits** are called the **Conjugacy Classes** of G .

Conjugacy Classes: Let G act on itself then $Conj(g) := \{h \cdot g | h \in G\} = \{hgh^{-1} | h \in G\}$

Conjugate: If $g, g' \in G$, there exists $h \in G$ s.t. $g' = hgh^{-1}$. We say that g, g' conjugate. (即在同一个 class/orbit)

$\cdot$ ps: when G is abelian, each element is its own conjugacy class.

$\cdot \{e\}$ is always a conjugacy class.

$\cdot$ **Heuristic (Hint):** 如果两个操作在几何上是对称的, 那么这两个操作 conjugate.

Centralizer: $C(g) := Stab_G(g) = \{h \in G | h \cdot g = g\} = \{h \in G | gh = hg\} \subseteq G$, under conjugacy action.

Important: $Conj(g) = Orb_G(g), |C(g)| = |Stab_G(G)|$ under conjugacy action.

Center: $C(G) := \{g \in G | gh = hg, \text{for all } h \in G\} = \cap_{g \in G} C(g)$ (即 conjugacy action 下的 kernel)

$\cdot G$ is abelian iff $C(G) = G$

Corollary: $C(g), C(G)$ are subgroups of G .

Corollary: $\{g\}$ is a conjugacy class iff $g \in C(G)$. Hence **$C(G)$ is the union of all 1-element conjugacy class.**

Theorem: Let G be a finite group with conjugacy classes: $Conj_1, Conj_2, \dots, Conj_n$, Let $Conj_1 = \{e\}$

1. If $g \in Conj_k$ then $|Conj_k||C(g)| = |G|$

2. **Class Equation:** $|G| = |Conj_1| + |Conj_2| + \dots + |Conj_n|$ ps: 因为 G 也是 set X , orbit 将 X partition.

Theorem: If G is a group

1. If $|G| = p^k$, p is prime, then $|C(G)| \geq p$ 即 G 里面至少有 p 个元素可以和所有元素有交换律.

2. Every group of order p^2 , (p prime) is abelian.

3.7 Cayley's Theorem

Conjugation Action Express in Cycle Notation: Let $\tau \in S_n, \sigma \in S_n, \sigma = (a_1 \dots a_n)(b_1 \dots b_n) \dots$ disjoint

Then: $\tau\sigma\tau^{-1} = (\tau(a_1) \dots \tau(a_n))(\tau(b_1) \dots \tau(b_n)) \dots$

Theorem: Two permutations $\in S_n$ are conjugate iff have the same cycle-shape.

Theorem: The If $\sigma \in S_n$ have cycle-shape: $(1^{a_1}, 2^{a_2}, \dots, n^{a_n})$ then # of $\sigma = \frac{n!}{a_1!a_2! \dots a_n! \cdot 1^{a_1} 2^{a_2} \dots n^{a_n}}$

Def of Bij(X): $Bij(X) := \{\text{bijections } X \rightarrow X\} = S_{|X|}$

Lemma: If G act on X , then $\forall g \in G, f_g : X \rightarrow X$ defined $x \rightarrow g \cdot x$ is bijection.

· ps: 即每一个 permutation 可以定义是 G (Group) 中的一个 g 对所有 X 中全部元素 action 所形成的 permutation.

Theorem: If G is group act on X , and $\phi : G \rightarrow S_{|X|} = Bij(X)$

1. Action of G on $X \Leftrightarrow$ A group homomorphism ϕ

ps: Lemma + 是同态

2. Action is faithful ($Kernel = \cap Stab = \{e\}$) iff ϕ is injective.

ps: 在这里的 ϕ 时, $Ker(\phi) = \{e\} \Leftrightarrow$ Kernel of action = $\{e\}$ (3.4 proposition of Ker 的推广).

3. Action is faithful $\Rightarrow G \cong Im(\phi)$ (ϕ isomorphism) ps: $Im(\phi) \leq Bij(X)$

ps: (3.4 proposition of Ker 的推广).

Cayley's Theorem: Every finite group is isomorphic to a subgroup of a symmetric group.

3.8 Alternating Group, Platonic Solids

Recall Permutation: $P = \prod_{1 \leq i < j \leq n} (x_i - x_j)$, Let $X = \{P, -P\}$. Then, S_n acts on X by $\sigma \cdot P = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)})$

· Odd when length of σ is even; Even when length of σ is odd;

Alternating Group: $A_n :=$ All even permutations of S_n . $|A_n| = \frac{|S_n|}{2} = \frac{n!}{2}$

Dual: 对偶图像 $sym(\Gamma)$ 相同 正十二面体 (Dodecahedron) 与正二十面体 (Icosahedron) 对偶

正四面体 (Tetrahedron) 自己对偶 正八面体 (Octahedron) 与正六面体 (Cube) 对偶.

3.9 Groups of Small Order

$ G $	1	2	3	4	5	6	7
$\cong$	$\{0\}$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_4, \mathbb{Z}_2 \times \mathbb{Z}_2$	$\mathbb{Z}_5$	$\mathbb{Z}_6, \mathbb{Z}_2 \times \mathbb{Z}_3, D_3$	$\mathbb{Z}_7$

3.10 Useful Graphs

